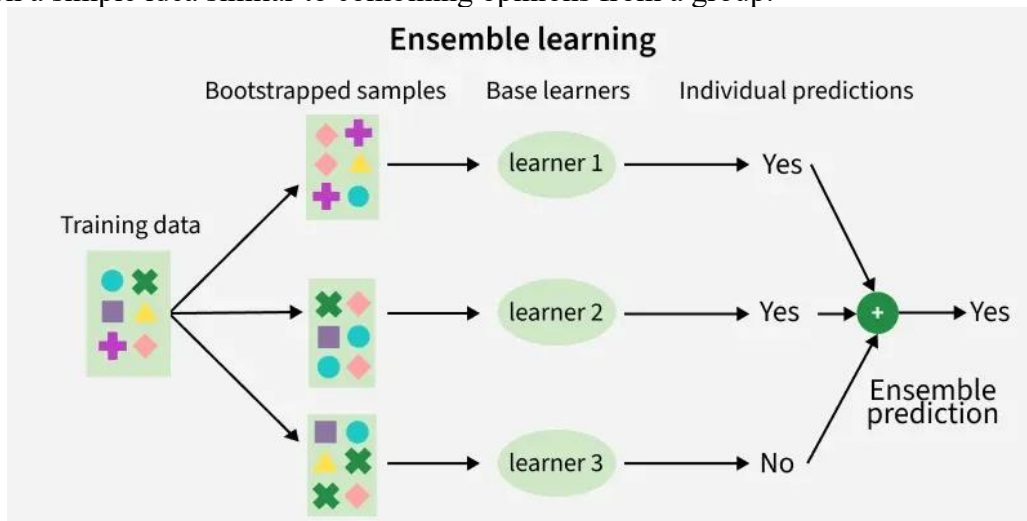


Q1. What is Ensemble Learning technique? What are the types of it?

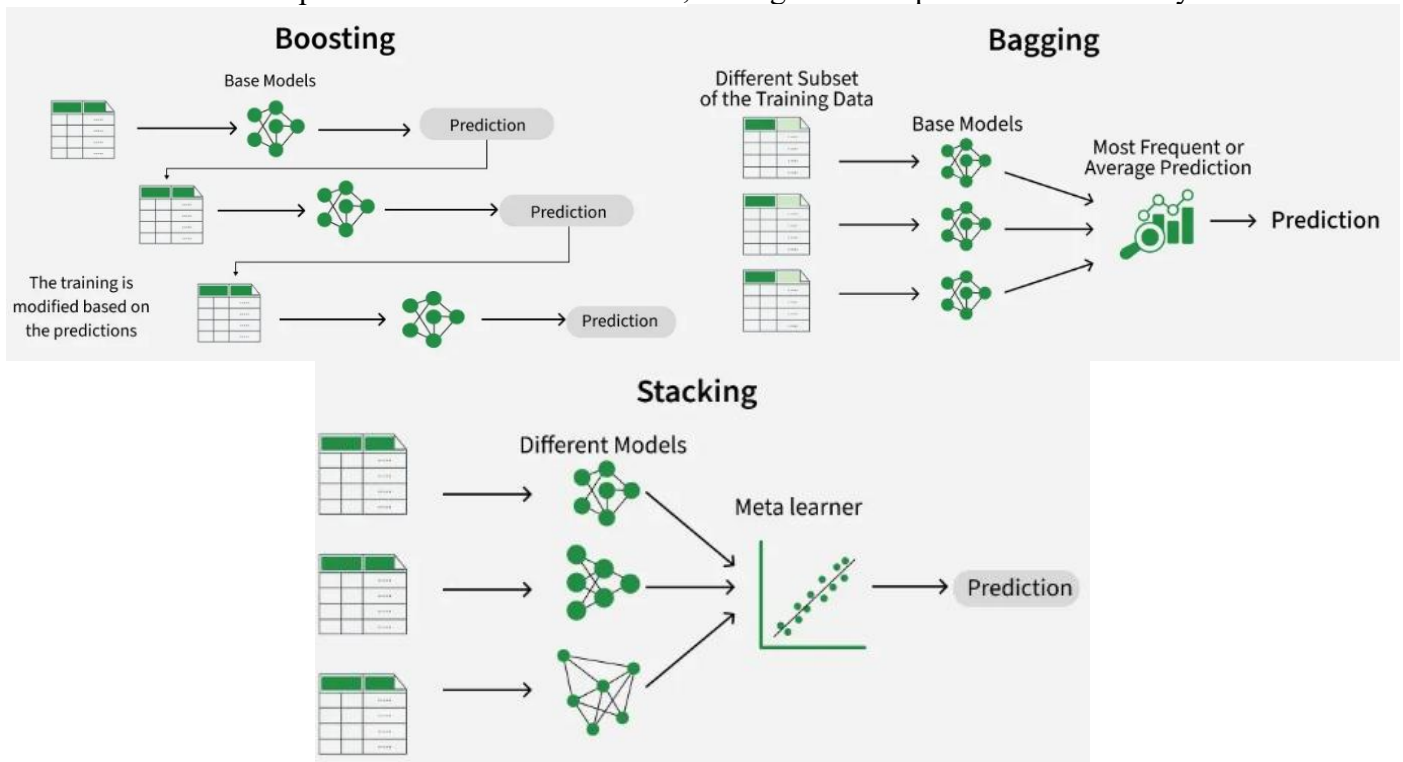
Ensemble learning is a method where multiple models are combined instead of using just one. Even if individual models are weak, combining their results gives more accurate and reliable predictions.

- Uses multiple models together to improve overall accuracy.
- Reduces errors by balancing mistakes across models.
- Works on a simple idea similar to combining opinions from a group.



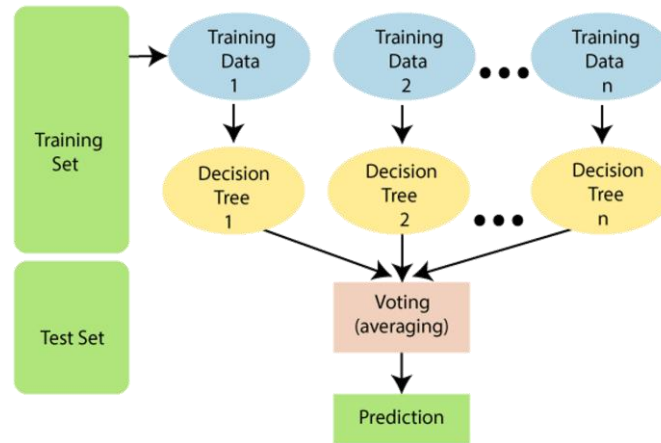
Types of Ensemble Learning

1. **Bagging (Bootstrap Aggregating):** Models are trained independently on different random subsets of the training data. Their results are then combined—usually by averaging (for regression) or voting (for classification). This helps reduce variance and prevents overfitting.
2. **Boosting:** Models are trained one after another. Each new model focuses on fixing the errors made by the previous ones. The final prediction is a weighted combination of all models, which helps reduce bias and improve accuracy.
3. **Stacking (Stacked Generalization):** Multiple different models (often of different types) are trained and their predictions are used as inputs to a final model, called a meta-model. The meta-model learns how to best combine the predictions of the base models, aiming for better performance than any individual model.



Q2. What is Random Forest Algorithm? How does it work? Explain with a real-life example.

Random Forest is a classifier that comprises a number of decision trees on various subsets of the provided dataset and takes the average to enhance the predicted accuracy of that dataset. Instead of depending on a single decision tree, the random forest collects the predictions from each tree and predicts the final output based on the majority vote of predictions.



Why use Random Forest?

Here are some reasons why we should utilize the Random Forest algorithm:

- It requires shorter training time than other algorithms.
- It predicts output with great accuracy, and it works efficiently even on big datasets.
- It can also retain accuracy when a significant amount of data is absent.

How does the Random Forest Algorithm Work?

Random Forest operates in two stages: the first is to generate the random forest by mixing N decision trees, and the second is to make predictions for each tree generated in the first phase.

Step 1: Choose K data points at random from the training set.

Step 2: Create decision trees for the specified data points (Subsets).

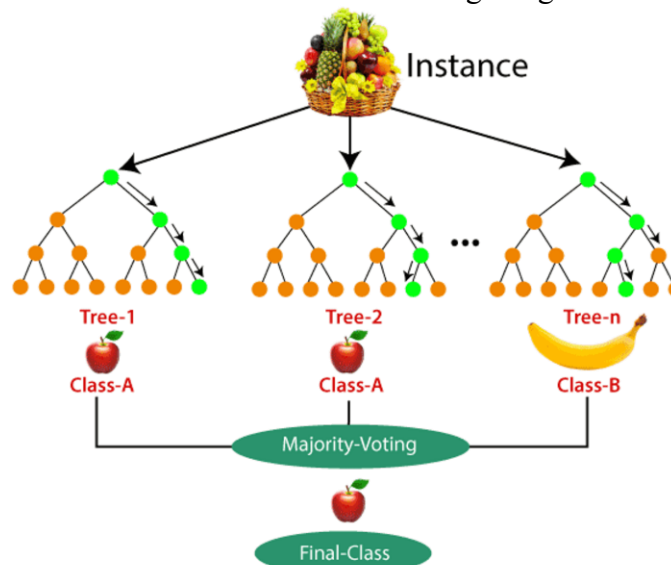
Step 3: Determine the number N for the number of decision trees you wish to construct.

Step 4: Repeat Steps 1 and 2.

Step 5: Find the predictions of each decision tree for new data points and assign the new data points to the category with the most votes.

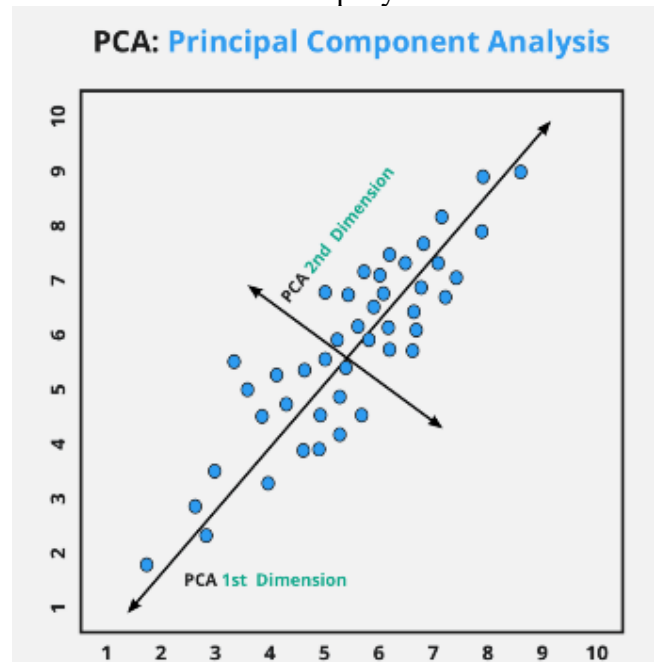
The following random forest algorithm example will help you understand how the algorithm works:

Assume there is a dataset containing several fruit photos. As a result, the Random Forest classifier is given this dataset. The dataset is subdivided and distributed to each decision tree. During the training phase, each decision tree gives a prediction result, and when a new data point occurs, the Random Forest classifier predicts the final choice based on the majority of outcomes. Consider the following image:



➤ Principal Component Analysis (PCA)

PCA (Principal Component Analysis) is a dimensionality reduction technique and helps us to reduce the number of features in a dataset while keeping the most important information. It changes complex datasets by transforming correlated features into a smaller set of uncorrelated components. PCA uses linear algebra to transform data into new features called principal components. It finds these by calculating eigenvectors (directions) and eigenvalues (importance) from the covariance matrix. PCA selects the top components with the highest eigenvalues and projects the data onto them to simplify the dataset.



PCA converts the original features into new features called **principal components**. These components are:

- **Uncorrelated**
- **Capture maximum variance**
- **Ordered** (PC1 captures the most variance, PC2 the second most, etc.)

PCA works by finding directions in the data where the data varies the most and projecting the data onto those directions.

🎯 When to Use PCA

1. To reduce dimensionality: Useful when you have many features and want to keep only the most important ones.
2. To avoid the curse of dimensionality: High-dimensional data can cause overfitting; PCA reduces dimensions safely.
3. To remove multicollinearity: If features are highly correlated, PCA combines them into fewer independent components. This helps linear models like logistic regression or linear regression.
4. For visualization: PCA can convert high-dimensional data into 2D or 3D for easy visualization.
5. To speed up machine learning models: Fewer features → faster training.
6. For noise reduction: PCA filters out noise by keeping only the components with high variance.

🔍 What is Curse of Dimensionality?

The **Curse of Dimensionality** refers to the problems that occur when the number of features (dimensions) in a dataset becomes very large.

As the number of dimensions increases:

- Data becomes sparse (spread out).
- Finding similar data points becomes difficult.
- Machine learning models require much more data and computation.
- Model accuracy often decreases.

➤ **Step to solve numerical on PCA:**

Step – 1: Find $\bar{x}$ and $\bar{y}$

Step – 2: calculate covariance matrix $(x - \bar{x})(x - \bar{x})$ using a simple table.

Step – 3: calculate covariance matrix $(y - \bar{y})(y - \bar{y})$ using a simple table.

Step – 4: calculate covariance matrix $(x - \bar{x})(y - \bar{y})$ using a simple table.

Step – 5: covariance matrix

$$cov(x, y) = \sum_{i=1}^n \frac{(x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Step – 6: covariance matrix

$$c = \begin{bmatrix} cov(x, x) & cov(x, y) \\ cov(y, x) & cov(y, y) \end{bmatrix}$$

Step – 7: find eigen values

$$C = \lambda I$$

Step – 8: find eigen vectors

$$CV = \lambda V$$

Step – 9: project the data of principle component on graph.

Q3. Apply PCA on the following data and find the first principal component.

X	2.5	0.5	2.2	1.9	3.1	2.3	2	1	1.5	1.1
Y	2.4	0.7	2.9	2.2	3	2.7	1.6	1.1	1.6	0.9

Solution:

Point	X	Y
1	2.5	2.4
2	0.5	0.7
3	2.2	2.9
4	1.9	2.2
5	3.1	3.0
6	2.3	2.7
7	2.0	1.6
8	1.0	1.1
9	1.5	1.6
10	1.1	0.9

Number of samples,

$$n = 10$$

Step 1: Find the Mean ($\bar{x}$ and $\bar{y}$)

Mean of X

$$\bar{x} = \frac{2.5 + 0.5 + 2.2 + 1.9 + 3.1 + 2.3 + 2 + 1 + 1.5 + 1.1}{10} = \frac{18.1}{10} = 1.81$$

Mean of Y

$$\bar{y} = \frac{2.4 + 0.7 + 2.9 + 2.2 + 3 + 2.7 + 1.6 + 1.1 + 1.6 + 0.9}{10} = \frac{19.1}{10} = 1.91$$

Therefore,

$$\bar{x} = 1.81, \bar{y} = 1.91$$

Step 2: Calculate Covariance Matrix (X, X)

First calculate

$$(x - \bar{x})$$

Then square it.

X	$X - \bar{x}$	$(x - \bar{x})^2$
2.5	0.69	0.4761
0.5	-1.31	1.7161
2.2	0.39	0.1521
1.9	0.09	0.0081
3.1	1.29	1.6641
2.3	0.49	0.2401
2.0	0.19	0.0361
1.0	-0.81	0.6561
1.5	-0.31	0.0961
1.1	-0.71	0.5041

Sum

$$\sum (x - \bar{x})^2 = 5.549$$

Now,

$$\text{cov}(x, x) = \frac{5.549}{9} = 0.6166$$

Step 3: Calculate Covariance Matrix (Y, Y)

Y	$Y - \bar{y}$	$((y - \bar{y})^2)$
2.4	0.49	0.2401
0.7	-1.21	1.4641
2.9	0.99	0.9801
2.2	0.29	0.0841
3.0	1.09	1.1881
2.7	0.79	0.6241
1.6	-0.31	0.0961
1.1	-0.81	0.6561
1.6	-0.31	0.0961
0.9	-1.01	1.0201

Sum

$$\sum (y - \bar{y})^2 = 6.449$$

Therefore,

$$\text{cov}(y, y) = \frac{6.449}{9} = 0.7166$$

Step 4: Calculate Covariance Matrix (X, Y)

Multiply

$$(x - \bar{x})(y - \bar{y})$$

Point	$(x - \bar{x})$	$(y - \bar{y})$	Product
1	0.69	0.49	0.3381
2	-1.31	-1.21	1.5851
3	0.39	0.99	0.3861
4	0.09	0.29	0.0261
5	1.29	1.09	1.4061
6	0.49	0.79	0.3871
7	0.19	-0.31	-0.0589
8	-0.81	-0.81	0.6561
9	-0.31	-0.31	0.0961
10	-0.71	-1.01	0.7171

Sum

$$\sum (x - \bar{x})(y - \bar{y}) = 5.539$$

Step 5: Find Covariance: Formula

$$\text{cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Substitute,

$$= \frac{5.539}{9}$$
$$\boxed{\text{cov}(x, y) = 0.6154}$$

Since covariance is symmetric,

$$\boxed{\text{cov}(y, x) = 0.6154}$$

Step 6: Covariance Matrix

$$C = \begin{bmatrix} \text{cov}(x, x) & \text{cov}(x, y) \\ \text{cov}(y, x) & \text{cov}(y, y) \end{bmatrix}$$

Substitute the values,

$$\boxed{C = \begin{bmatrix} 0.6166 & 0.6154 \\ 0.6154 & 0.7166 \end{bmatrix}}$$

Step 7: Find Eigenvalues: The eigenvalues are found by solving

$$|C - \lambda I| = 0$$

Where, C = covariance matrix, I = Identity matrix

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Multiply λ with the identity matrix

$$\lambda I = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

Subtract λI from C

$$\begin{vmatrix} 0.6166 - \lambda & 0.6154 \\ 0.6154 & 0.7166 - \lambda \end{vmatrix} = 0$$

Expand,

$$(0.6166 - \lambda)(0.7166 - \lambda) - (0.6154)^2 = 0$$

This gives

$$\lambda^2 - 1.3332\lambda + 0.0626 = 0$$

Solve the quadratic equation:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Here,

$$a = 1, b = -1.3332, c = 0.0631$$

Substituting,

$$\lambda = \frac{1.3332 \pm \sqrt{(1.3332)^2 - 4(1)(0.0631)}}{2}$$

After calculation,

$$\boxed{\lambda_1 = 1.284}, \boxed{\lambda_2 = 0.049}$$

The larger eigenvalue is

$$\boxed{\lambda_1 = 1.284}$$

This means **the first principal component contains the maximum variance**, so we will use it as the new axis.

Step 8: Find Eigenvector

Solve

$$CV = \lambda V$$

Move everything to one side

$$(C - \lambda I)V = 0$$

Since

$$\lambda = 1.284$$

we have

$$\begin{aligned} C - \lambda I &= \begin{bmatrix} 0.6166 - 1.284 & 0.6154 \\ 0.6154 & 0.7166 - 1.284 \end{bmatrix} \\ &= \begin{bmatrix} -0.6674 & 0.6154 \\ 0.6154 & -0.5674 \end{bmatrix} \end{aligned}$$

Multiply with the unknown vector

$$\begin{bmatrix} -0.6674 & 0.6154 \\ 0.6154 & -0.5674 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives two equations:

Equation (1)

$$-0.6674v_1 + 0.6154v_2 = 0$$

Equation (2)

$$0.6154v_1 - 0.5674v_2 = 0$$

Both equations represent the same line, so solving either one is sufficient.

$$0.6154v_2 = 0.6674v_1$$

Divide both sides by 0.6154:

$$v_2 = \frac{0.6674}{0.6154}v_1$$

$$v_2 = 1.0845v_1$$

Assume

$$v_1 = 1$$

Then

$$v_2 = 1.0845$$

So, the (unnormalized) eigenvector is

$$\begin{bmatrix} 1 \\ 1.0845 \end{bmatrix}$$

Normalize the vector

PCA uses unit vectors, meaning the vector length should be 1.

Compute its length:

$$\sqrt{1^2 + 1.0845^2} = \sqrt{2.176} \approx 1.475$$

Divide each component by 1.475:

$$\begin{aligned} v_1 &= \frac{1}{1.475} = 0.677 \\ v_2 &= \frac{1.0845}{1.475} = 0.736 \end{aligned}$$

Therefore,

$$V = \begin{bmatrix} 0.677 \\ 0.736 \end{bmatrix}$$

Are the eigen vectors.

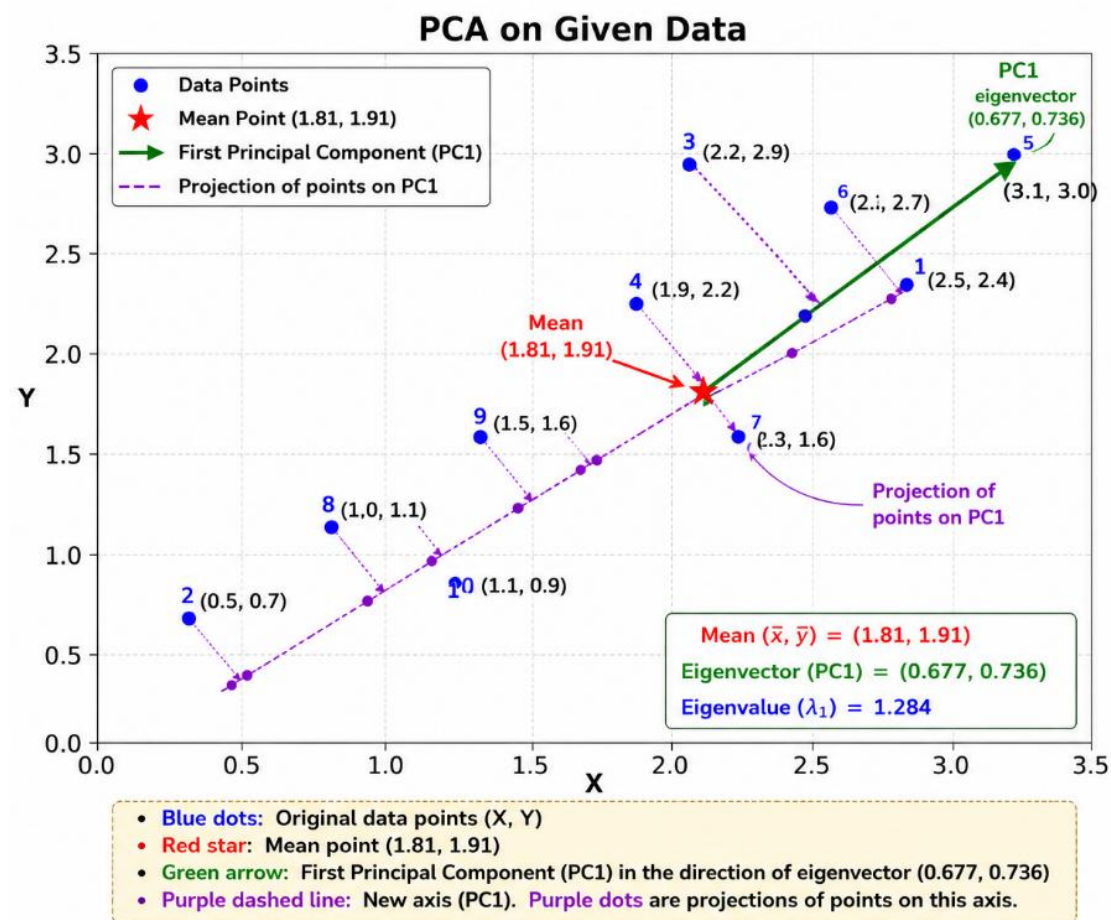
Step 9: Project Data on the Principal Component

The projection formula is

$$z = (x - \bar{x}) \times 0.677 + (y - \bar{y}) \times 0.736$$

Let's calculate all the points.

Point	X	Y	$x - \bar{x}$	$y - \bar{y}$	Projection (z)
1	2.5	2.4	0.69	0.49	$(0.69)(0.677) + (0.49)(0.736) = 0.828$
2	0.5	0.7	-1.31	-1.21	$(-1.31)(0.677) + (-1.21)(0.736) = -1.778$
3	2.2	2.9	0.39	0.99	$(0.39)(0.677) + (0.99)(0.736) = 0.993$
4	1.9	2.2	0.09	0.29	$0.09 \times 0.677 + 0.29 \times 0.736 = 0.274$
5	3.1	3.0	1.29	1.09	$1.29 \times 0.677 + 1.09 \times 0.736 = 1.677$
6	2.3	2.7	0.49	0.79	$0.49 \times 0.677 + 0.79 \times 0.736 = 0.913$
7	2.0	1.6	0.19	-0.31	$0.19 \times 0.677 - 0.31 \times 0.736 = -0.099$
8	1.0	1.1	-0.81	-0.81	$-0.81 \times 0.677 - 0.81 \times 0.736 = -1.145$
9	1.5	1.6	-0.31	-0.31	$-0.31 \times 0.677 - 0.31 \times 0.736 = -0.438$
10	1.1	0.9	-0.71	-1.01	$-0.71 \times 0.677 - 1.01 \times 0.736 = -1.224$



Q4. Apply PCA on the following data and find the first principal component.

X	2	3	4	5	6	7
Y	1	5	3	6	7	8

Solution:

Point	X	Y
1	2	1
2	3	5
3	4	3
4	5	6
5	6	7
6	7	8

Number of samples,

$$n = 6$$

Step 1: Find the Mean $\bar{x}$ and $\bar{y}$

Mean of X

$$\begin{aligned}\bar{x} &= \frac{2 + 3 + 4 + 5 + 6 + 7}{6} \\ &= \frac{27}{6} = 4.5\end{aligned}$$

Mean of Y

$$\begin{aligned}\bar{y} &= \frac{1 + 5 + 3 + 6 + 7 + 8}{6} \\ &= \frac{30}{6} = 5\end{aligned}$$

Therefore,

$$\boxed{\bar{x} = 4.5, \bar{y} = 5}$$

Step 2: Calculate Covariance Matrix (X, X)

Calculate

$$(x - \bar{x})^2$$

X	$(x - \bar{x})$	$(x - \bar{x})^2$
2	-2.5	6.25
3	-1.5	2.25
4	-0.5	0.25
5	0.5	0.25
6	1.5	2.25
7	2.5	6.25

Sum,

$$\sum (x - \bar{x})^2 = 17.5$$

Now,

$$\text{cov}(x, x) = \frac{17.5}{6 - 1} = \frac{17.5}{5} = 3.5$$

Step 3: Calculate Covariance Matrix (Y, Y)

Y	$(y - \bar{y})$	
1	-4	16
5	0	0
3	-2	4
6	1	1
7	2	4
8	3	9

Sum,

$$16 + 0 + 4 + 1 + 4 + 9 = 34$$

Therefore,

$$\text{cov}(y, y) = \frac{34}{5} = 6.8$$

Step 4: Calculate Covariance Matrix (X, Y)

Find

$$(x - \bar{x})(y - \bar{y})$$

Point	$(x - \bar{x})$	$(y - \bar{y})$	Product
1	-2.5	-4	10
2	-1.5	0	0
3	-0.5	-2	1
4	0.5	1	0.5
5	1.5	2	3
6	2.5	3	7.5

Sum,

$$10 + 0 + 1 + 0.5 + 3 + 7.5 = 22$$

Step 5: Find Covariance

Using

$$\text{cov}(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n - 1}$$

Substitute,

$$= \frac{22}{5} = 4.4$$

Since covariance is symmetric,

$$\text{cov}(x, y) = \text{cov}(y, x) = 4.4$$

Step 6: Construct the Covariance Matrix

$$C = \begin{bmatrix} \text{cov}(x, x) & \text{cov}(x, y) \\ \text{cov}(y, x) & \text{cov}(y, y) \end{bmatrix}$$

Substituting the values,

$$C = \begin{bmatrix} 3.5 & 4.4 \\ 4.4 & 6.8 \end{bmatrix}$$

Step 7: Find the Eigenvalues

We solve

$$|C - \lambda I| = 0$$

Substitute the covariance matrix,

$$\begin{vmatrix} 3.5 - \lambda & 4.4 \\ 4.4 & 6.8 - \lambda \end{vmatrix} = 0$$

Using

$$ad - bc$$

we get

$$(3.5 - \lambda)(6.8 - \lambda) - 4.4^2 = 0$$

Expand,

$$23.8 - 10.3\lambda + \lambda^2 - 19.36 = 0$$

Simplify,

$$\lambda^2 - 10.3\lambda + 4.44 = 0$$

Using the quadratic formula,

$$\begin{aligned} \lambda &= \frac{10.3 \pm \sqrt{10.3^2 - 4(4.44)}}{2} \\ &= \frac{10.3 \pm 9.40}{2} \end{aligned}$$

Hence,

$$\lambda_1 \approx 9.85, \lambda_2 \approx 0.45$$

The larger eigenvalue is

$$\lambda_1 = 9.85$$

Step 8: Find the Eigenvector (First Principal Component)

Use

$$(C - \lambda I)V = 0$$

For

$$\begin{bmatrix} 3.5 - 9.85 & 4.4 \\ 4.4 & 6.8 - 9.85 \end{bmatrix} = \begin{bmatrix} -6.35 & 4.4 \\ 4.4 & -3.05 \end{bmatrix}$$

Let

$$V = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

Then,

$$-6.35v_1 + 4.4v_2 = 0$$

Therefore,

$$\begin{aligned} v_2 &= \frac{6.35}{4.4} v_1 \\ v_2 &= 1.443v_1 \end{aligned}$$

Assume

$$v_1 = 1$$

Then

$$v_2 = 1.443$$

The unnormalized eigenvector is

$$\begin{bmatrix} 1 \\ 1.443 \end{bmatrix}$$

Normalize it.

Length,

$$\sqrt{1^2 + 1.443^2} = \sqrt{3.082} = 1.756$$

Divide each value by 1.756,

$$v_1 = \frac{1}{1.756} = 0.569$$

$$v_2 = \frac{1.443}{1.756} = 0.822$$

Therefore,

$$V = \begin{bmatrix} 0.569 \\ 0.822 \end{bmatrix}$$

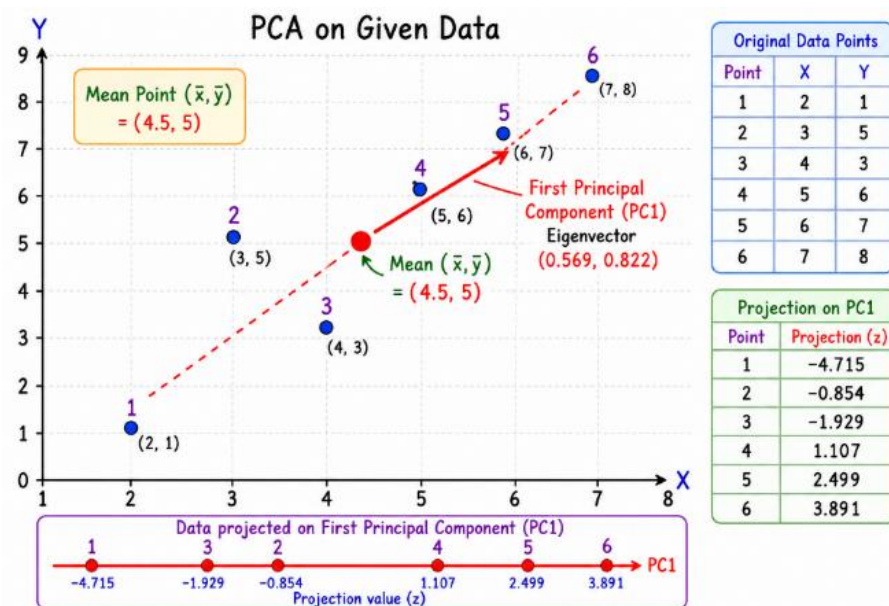
This is the first principal component (PC1).

Step 9: Project the Data onto PC1

The projection formula is

$$z = (x - \bar{x})(0.569) + (y - \bar{y})(0.822)$$

Point	$(x - \bar{x})$	$(y - \bar{y})$	Projection (z)
1	-2.5	-4	-4.715
2	-1.5	0	-0.854
3	-0.5	-2	-1.929
4	0.5	1	1.107
5	1.5	2	2.499
6	2.5	3	3.891



Bayes Theorem

Bayes's Theorem is a probability rule that helps us calculate the chance of an event by using both prior knowledge and new information. In simple terms, it allows us to update our understanding or prediction when new evidence becomes available. It is widely used in statistics, machine learning, and decision-making. The figure below shows the formula of Bayes's Theorem, along with an explanation of each probability term used in the formula:

The diagram illustrates Bayes' Theorem and its components. On the left, a green box contains the formula $P(A|B) = \frac{P(A) P(B|A)}{P(B)}$. To the right of this box, four definitions are listed: $P(A)$ = The probability of A occurring, $P(B)$ = The probability of B occurring, $P(A|B)$ = The probability of A given B, and $P(B|A)$ = The probability of B given A. Further right, the formula $P(A|B) = P(A) \times \frac{P(B|A)}{P(B)}$ is shown with colored boxes and labels: $P(A|B)$ is in a pink box labeled 'posterior', $P(A)$ is in a cyan box labeled 'prior', $P(B|A)$ is in a green box labeled 'likelihood', and $P(B)$ is in a blue box labeled 'marginal'.

$$P(A|B) = \frac{P(A) P(B|A)}{P(B)}$$

$P(A)$ = The probability of A occurring
 $P(B)$ = The probability of B occurring
 $P(A|B)$ = The probability of A given B
 $P(B|A)$ = The probability of B given A

$$P(A|B) = P(A) \times \frac{P(B|A)}{P(B)}$$

posterior prior likelihood marginal

Bayes' Theorem provides a principled way to reverse conditional probabilities. It is defined as:

$$P(y | X) = \frac{P(X | y) \cdot P(y)}{P(X)}$$

Where:

- $P(y | X)$:Posterior probability, probability of class y given features X
- $P(X | y)$:Likelihood, probability of features X given class y OR Conditional Probability
- $P(y)$:Prior probability of class y
- $P(X)$:Marginal likelihood or evidence

Naive Bayes Working:

1. The Naive Assumption

The "naive" in Naive Bayes comes from the assumption that all features are independent given the class. That is:

$$P(x_1, x_2, \dots, x_n | y) = P(x_1 | y) \cdot P(x_2 | y) \cdots P(x_n | y)$$

Thus, Bayes' theorem becomes:

$$P(y | x_1, \dots, x_n) = \frac{P(y) \cdot \prod_{i=1}^n P(x_i | y)}{P(X)}$$

Since the denominator is constant for a given input, we can write:

$$P(y | x_1, \dots, x_n) \propto P(y) \cdot \prod_{i=1}^n P(x_i | y)$$

2. Constructing the Naive Bayes Classifier

We compute the posterior for each class y and choose the class with the highest probability:

$$\hat{y} = \arg \max_y P(y) \cdot \prod_{i=1}^n P(x_i | y)$$

This becomes our Naive Bayes classifier.

Q5. Apply Naïve Bayes Classifier on the following data and predict the result for the new input data.

Applicant	Income	Credit Score	Employment	Loan Status
A1	High	Good	Yes	Approved
A2	High	Poor	Yes	Approved
A3	Low	Good	Yes	Approved
A4	Low	Poor	No	Rejected
A5	High	Poor	No	Rejected
A6	Low	Good	No	Rejected

Predict the class of a new applicant with:

- **Income = High**
- **Credit Score = Good**
- **Employment = No**

Solution

Step 1: Prior Probability

Total applicants = 6

Approved = 3

Rejected = 3

$$P(\text{Approved}) = \frac{3}{6} = 0.5$$
$$P(\text{Rejected}) = \frac{3}{6} = 0.5$$

Step 2: Likelihood:

For Approved

Income	Credit Score	Employment
High	Good	Yes
High	Poor	Yes
Low	Good	Yes

$$P(\text{High} | \text{Approved}) = \frac{2}{3}$$

$$P(\text{Good} | \text{Approved}) = \frac{2}{3}$$

$$P(\text{No} | \text{Approved}) = \frac{0}{3} = 0$$

For Rejected

Income	Credit Score	Employment
Low	Poor	No
High	Poor	No
Low	Good	No

$$P(\text{High} | \text{Rejected}) = \frac{1}{3}$$

$$P(\text{Good} | \text{Rejected}) = \frac{1}{3}$$

$$P(\text{No} | \text{Rejected}) = \frac{3}{3} = 1$$

Step 3: Naive Bayes classifier:

Approved Score

$$0.5 \times \frac{2}{3} \times \frac{2}{3} \times 0 = 0$$

Rejected Score

$$0.5 \times \frac{1}{3} \times \frac{1}{3} \times 1 = \frac{1}{18} \approx 0.056$$

Since

$$0.056 > 0$$

Predicted Class = Rejected

Q6. Apply Naïve Bayes Classifier on the following data and predict the result for the new input data.

Email	Contains "Offer"	Contains "Free"	Class
E1	Yes	Yes	Spam
E2	Yes	No	Spam
E3	No	Yes	Spam
E4	No	No	Not Spam
E5	No	No	Not Spam
E6	Yes	No	Not Spam

Using the **Naive Bayes Classifier**, predict the class of a new email having:

- Offer = Yes
- Free = Yes

Solution

Step 1: Calculate Prior Probabilities

Total emails = 6

Spam emails = 3

Not Spam emails = 3

$$P(\text{Spam}) = \frac{3}{6} = 0.5$$

$$P(\text{Not Spam}) = \frac{3}{6} = 0.5$$

Step 2: Calculate Conditional Probabilities

For Spam

Spam emails:

Offer	Free
Yes	Yes
Yes	No
No	Yes

Probability of Offer = Yes

2 out of 3 Spam emails contain Offer.

$$P(\text{Offer}=\text{Yes} \mid \text{Spam}) = \frac{2}{3}$$

Probability of Free = Yes

2 out of 3 Spam emails contain Free.

$$P(\text{Free}=\text{Yes} \mid \text{Spam}) = \frac{2}{3}$$

For Not Spam

Not Spam emails:

Offer	Free
No	No
No	No
Yes	No

Probability of Offer = Yes

$$P(\text{Offer}=\text{Yes} \mid \text{Not Spam}) = \frac{1}{3}$$

Probability of Free = Yes

No Not Spam email contains Free.

$$P(\text{Free}=\text{Yes} \mid \text{Not Spam}) = \frac{0}{3} = 0$$

Step 3: Apply Naive Bayes Classifier

Spam Score

$$P(\text{Spam}) \times P(\text{Offer}=\text{Yes} \mid \text{Spam}) \times P(\text{Free}=\text{Yes} \mid \text{Spam})$$

$$= 0.5 \times \frac{2}{3} \times \frac{2}{3}$$

$$= 0.5 \times \frac{4}{9} = \frac{2}{9} \approx 0.222$$

Not Spam Score

$$0.5 \times \frac{1}{3} \times 0 = 0$$

Since

$$0.222 > 0$$

the email is classified as:

Predicted Class = Spam

Q7. Apply Naïve Bayes Classifier on the following data and predict the result for the new input data.

Email	Offer	Free	Urgent	Class
E1	Yes	Yes	Yes	Spam
E2	Yes	Yes	No	Spam
E3	Yes	No	Yes	Spam
E4	No	No	No	Not Spam
E5	No	Yes	No	Not Spam
E6	Yes	No	No	Not Spam

Using the **Naïve Bayes Classifier**, predict the class of a new email with:

- Offer = Yes
- Free = Yes
- Urgent = Yes

Solution

Step 1: Calculate Prior Probabilities

Total emails = 6

Spam = 3

Not Spam = 3

$$P(\text{Spam}) = \frac{3}{6} = 0.5$$

$$P(\text{Not Spam}) = \frac{3}{6} = 0.5$$

Step 2: Calculate Conditional Probabilities

For Spam

Spam emails:

Offer	Free	Urgent
Yes	Yes	Yes
Yes	Yes	No
Yes	No	Yes

Offer = Yes: All 3 Spam emails contain Offer.

$$P(\text{Offer}=\text{Yes} \mid \text{Spam}) = \frac{3}{3} = 1$$

Free = Yes: 2 Spam emails contain Free.

$$P(\text{Free}=\text{Yes} \mid \text{Spam}) = \frac{2}{3}$$

Urgent = Yes: 2 Spam emails contain Urgent.

$$P(\text{Urgent}=\text{Yes} \mid \text{Spam}) = \frac{2}{3}$$

For Not Spam

Not Spam emails:

Offer	Free	Urgent
No	No	No
No	Yes	No
Yes	No	No

Offer = Yes

$$P(\text{Offer}=\text{Yes} \mid \text{Not Spam}) = \frac{1}{3}$$

Free = Yes

$$P(\text{Free}=\text{Yes} \mid \text{Not Spam}) = \frac{1}{3}$$

Urgent = Yes: No Not Spam email contains Urgent.

$$P(\text{Urgent}=\text{Yes} \mid \text{Not Spam}) = 0$$

Step 3: Apply Naive Bayes Classifier

Spam Score

$$0.5 \times 1 \times \frac{2}{3} \times \frac{2}{3}$$

$$= 0.5 \times \frac{4}{9} = \frac{2}{9} \approx 0.222$$

Not Spam Score

$$0.5 \times \frac{1}{3} \times \frac{1}{3} \times 0 = 0$$

Since

$$0.222 > 0$$

the new email is classified as:

Predicted Class = Spam

Q8. Apply Naïve Bayes Classifier on the following data and predict the result for the new input data.

Day	Outlook	Temperature	Humidity	Windy	Play Golf
0	Rainy	Hot	High	False	Yes
1	Rainy	Hot	High	True	No
2	Overcast	Hot	High	False	Yes
3	Sunny	Mild	High	False	No
4	Sunny	Cool	Normal	False	Yes
5	Sunny	Cool	Normal	True	No
6	Overcast	Cool	Normal	True	Yes
7	Rainy	Mild	High	False	No
8	Rainy	Cool	Normal	False	Yes
9	Sunny	Mild	Normal	False	Yes
10	Rainy	Mild	Normal	True	Yes
11	Overcast	Mild	High	True	Yes
12	Overcast	Hot	Normal	False	Yes
13	Sunny	Mild	High	True	No

Using the **Naive Bayes Classifier**, predict the class of a new data with:

- Outlook = Sunny
- Temperature = Cool
- Humidity = High
- Windy = False

Solution:

Step 1: Prior Probability

$$\begin{aligned} \text{Total records} &= 14 \\ \text{Play Golf} = \text{Yes} &= 9 & P(\text{Yes}) &= \frac{9}{14} \\ \text{Play Golf} = \text{No} &= 5 & P(\text{No}) &= \frac{5}{14} \end{aligned}$$

Step 2: Conditional Probabilities

For Play Golf = Yes, there are **9** Yes records.

Outlook = Sunny: Sunny & Yes = 3

$$P(\text{Sunny} | \text{Yes}) = \frac{3}{9} = \frac{1}{3}$$

Temperature = Cool: Cool & Yes = 3

$$P(\text{Cool} | \text{Yes}) = \frac{3}{9} = \frac{1}{3}$$

Humidity = High: High & Yes = 3

$$P(\text{High} | \text{Yes}) = \frac{3}{9} = \frac{1}{3}$$

Windy = False: False & Yes = 6

$$P(\text{False} | \text{Yes}) = \frac{6}{9} = \frac{2}{3}$$

For Play Golf = No, there are **5** No records.

Outlook = Sunny: Sunny & No = 3

$$P(\text{Sunny} | \text{No}) = \frac{3}{5}$$

Temperature = Cool: Cool & No = 1

$$P(\text{Cool} | \text{No}) = \frac{1}{5}$$

Humidity = High: High & No = 4

$$P(\text{High} | \text{No}) = \frac{4}{5}$$

Windy = False: False & No = 2

$$P(\text{False} | \text{No}) = \frac{2}{5}$$

Step 3: Apply Naive Bayes Classifier

$$\text{Score}(\text{Yes}) = \frac{9}{14} \times \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} \times \frac{2}{3} = \frac{2}{189} \approx 0.0106$$

$$\text{Score}(\text{No}) = \frac{5}{14} \times \frac{3}{5} \times \frac{1}{5} \times \frac{4}{5} \times \frac{2}{5} = \frac{12}{1750} \approx 0.00686$$

Since

$$0.0106 > 0.00686$$

Final Answer

Predicted Class = Play Golf = Yes